

United International University

Department of Computer Science and Engineering

Course: *Electronics Lab [2124]*

PROJECT PROPOSAL

Smart Helmet Safety and Rider Monitoring System

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Abstract

Road traffic injuries remain one of the leading causes of preventable death among young riders, and a large share of these incidents involve motorcycles operated without a helmet, under the influence of alcohol, at unsafe speed, or by a fatigued rider. This proposal presents the design of an AI-Based Smart Motorcycle Rider Safety, Monitoring, Accident Detection and Anti-Theft System, a microcontroller-based prototype that combines camera-based image processing, environmental and motion sensing, biometric authentication, and GSM/GPS communication to address these risk factors in a single integrated unit. The system verifies that the rider is wearing a helmet before permitting the ignition to engage, screens for alcohol consumption, authenticates the rider through a fingerprint sensor, and continuously monitors speed, ambient light, and eye closure while the motorcycle is in motion. In the event of a probable accident, the system determines the rider's GPS location and transmits an emergency notification through a GSM module to a predefined contact, aiming to reduce the time between an incident and the arrival of help. This document outlines the project objectives, the hardware components selected for the prototype, the proposed system architecture, the development methodology, and the expected outcomes and limitations of the design.

1. Introduction

Motorcycles are the most common mode of personal transport in many densely populated regions because they are affordable and easy to maneuver through traffic. This convenience, however, comes with a disproportionately high accident and fatality rate compared to other vehicle types. Common contributing factors include riding without a helmet, riding under the influence of alcohol, excessive speed, rider fatigue, and delayed emergency response after a crash. Motorcycle theft is a further recurring problem, particularly for unattended vehicles that lack any active security mechanism beyond a mechanical lock.

This project proposes a single embedded system that addresses several of these problems together rather than in isolation. Instead of a device that only tracks location or only checks for a helmet, the proposed system links helmet detection, alcohol screening, rider authentication, speed monitoring, drowsiness detection, accident detection, and anti-theft protection into one coordinated unit built around an ESP32-based hardware platform. Each function relies on hardware that is realistic for a university-level prototype, and the system is designed so that unsafe conditions — no helmet, detected alcohol, or an unauthorized user — can prevent the motorcycle from starting in the first place, rather than only warning the rider after the fact.

The remainder of this document describes the objectives of the project, the individual features and the hardware components selected to implement them, the proposed system architecture, the development methodology, and the expected outcomes, limitations, estimated budget, and timeline for the prototype.

2. Objectives

The specific objectives of this project are:

- To design a camera-based helmet detection mechanism using the ESP32-CAM and OV2640 sensor that verifies proper helmet usage before the motorcycle ignition is enabled.
- To implement an alcohol-detection interlock using the MQ-3 gas sensor that prevents the motorcycle from starting when the rider's breath alcohol reading exceeds a predefined threshold.
- To develop a fingerprint-based rider authentication system so that only registered users can operate the motorcycle, and to raise an alert when an unregistered user attempts access.
- To build a real-time speed monitoring feature using a Hall effect speed sensor that warns the rider when a predefined speed limit is exceeded.

- To design an automatic low-light detection mechanism using an LDR sensor that adjusts the helmet/headlight illumination according to ambient light conditions.
- To implement an accident detection mechanism using motion and vibration sensors (MPU6050 and SW-420) capable of distinguishing a probable crash from normal riding vibration.
- To integrate GPS and GSM modules so that, upon accident detection, the rider's location is obtained and transmitted as an emergency SMS notification to a predefined contact.
- To design a drowsiness detection feature that monitors the rider's eye state through the onboard camera and issues a voice alert when signs of fatigue are observed.
- To provide anti-theft protection by combining fingerprint authentication with a buzzer and GSM-based alert whenever unauthorized access is attempted.
- To integrate all of the above sensors and modules around a single ESP32 DevKit controller into one coherent, testable prototype.

3. Proposed System and Key Features

The proposed system is organized around four functional groups: (1) helmet and rider detection, (2) rider safety and monitoring, (3) accident detection and emergency response, and (4) drowsiness detection with voice alerts. Each group is described below along with the hardware it depends on; a full account of every component is provided separately in Section 5.

3.1 AI-Based Helmet and Rider Detection

Before the ignition circuit is enabled, the ESP32-CAM module, fitted with an OV2640 camera, captures an image of the rider. Onboard or edge image-processing logic checks the captured frame for the presence of a helmet on the rider's head. The same frame can also be used to get a rough estimate of how many people are seated on the motorcycle, since more than one rider changes the safety requirements of the trip. Where the image is ambiguous — for example, a helmet-shaped object that is not actually worn correctly — the system is designed to treat the case conservatively and withhold ignition permission rather than assume compliance.

If no helmet is detected in the captured frame, the ESP32 DevKit withholds the signal that would normally energize the relay module connected to the ignition circuit, so the motorcycle cannot be started. Once a helmet is confirmed, this check hands control over to the alcohol-detection stage described next.

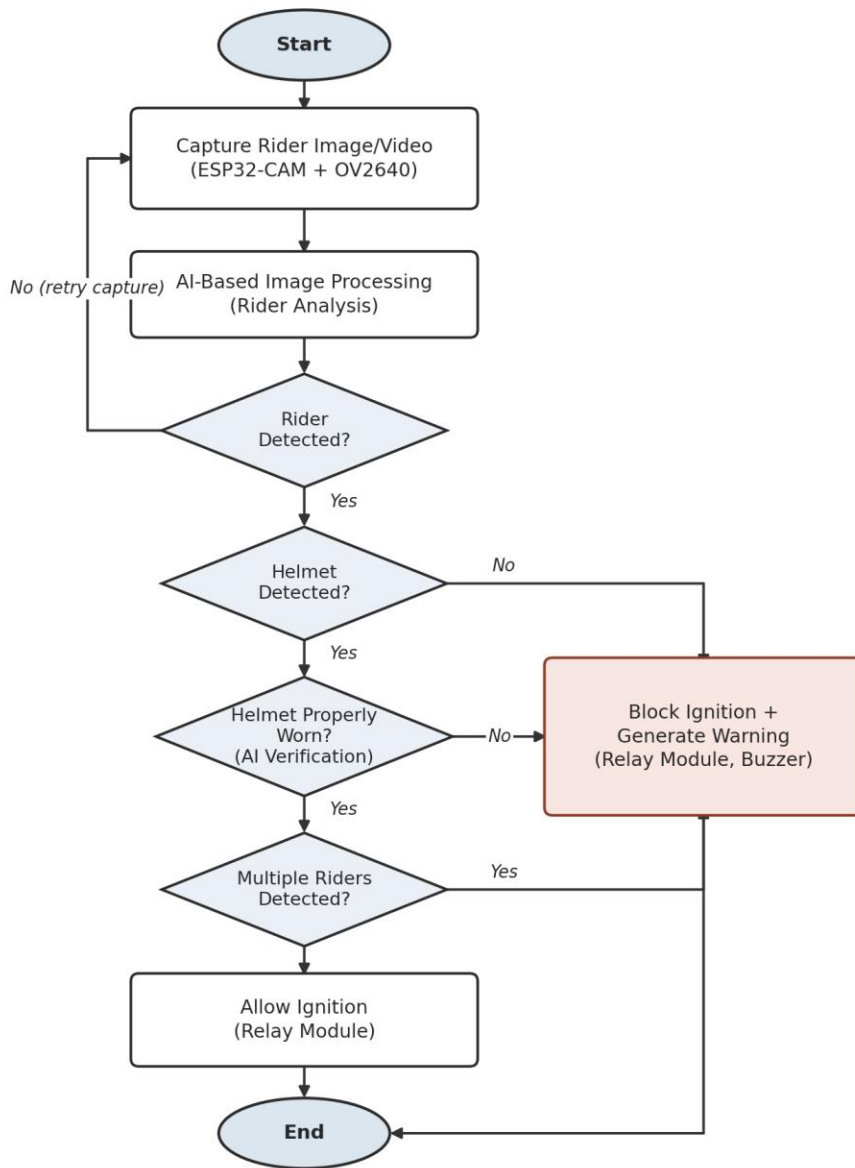


Figure 1: Flowchart of AI-Based Helmet and Rider Detection

3.2 Intelligent Rider Safety and Monitoring System

This group covers four related checks that run either before the motorcycle starts or continuously while it is in motion:

Alcohol screening: the MQ-3 sensor is warmed up and read shortly after the rider is authenticated. If the analog reading corresponds to a breath-alcohol level above the configured threshold, the ESP32 keeps the ignition relay open and triggers the buzzer as a warning.

Overspeed warning: while riding, the Hall effect speed sensor feeds pulse data to the ESP32, which converts pulse frequency into an approximate speed value. If this value exceeds a rider-configured limit, the system issues an audible warning through the buzzer or a short voice prompt through the speaker.

Automatic low-light adjustment: the LDR module continuously reports ambient light levels. When readings fall below a set threshold — for example, at dusk or inside a tunnel — the ESP32 increases the brightness of the connected LED/headlight circuit to improve visibility for the rider and other road users.

Fingerprint authentication and anti-theft alert: the fingerprint sensor is used both to authenticate an authorized rider before ignition and, when an unrecognized fingerprint is repeatedly presented, to treat the situation as a possible theft attempt. In that case, the buzzer sounds locally and the SIM800L GSM module sends an SMS alert to the owner's registered number.

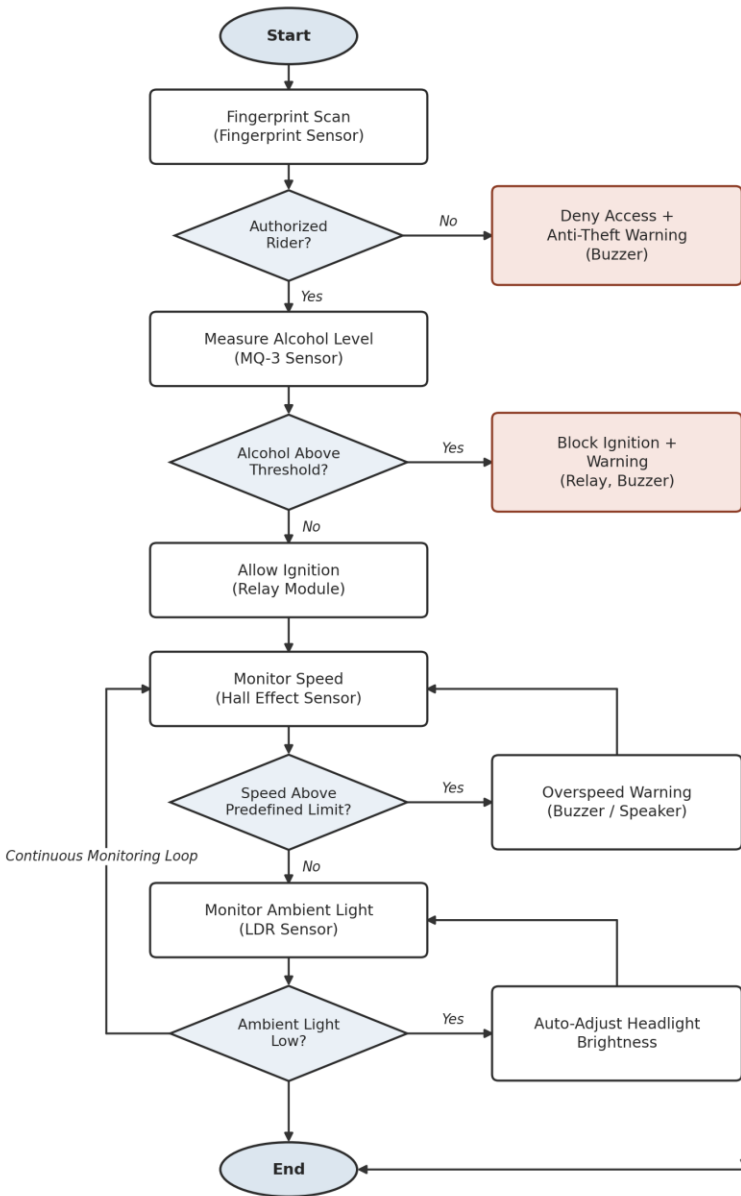


Figure 2: Flowchart of Intelligent Rider Safety and Monitoring System

3.3 Accident Detection and Emergency Response

Accident detection relies on the combination of the MPU6050 inertial measurement unit and the SW-420 vibration sensor. The MPU6050 continuously reports acceleration and angular-rate data, which the ESP32 uses to look for a signature consistent with a fall or sudden impact — a sharp deceleration or an abnormal change in orientation, for instance. The SW-420 provides a secondary, threshold-based vibration/shock indication that supports this decision and helps reduce false triggers from ordinary road bumps.

When both indicators point toward a probable accident, the ESP32-CAM can capture a still image of the scene for later reference, the NEO-6M GPS module is queried for the current coordinates, and the SIM800L GSM module sends an SMS containing the location link to a predefined emergency contact. This sequence is intended to shorten the time between a crash and the arrival of assistance, which is one of the most significant factors in survivable but time-critical injuries.

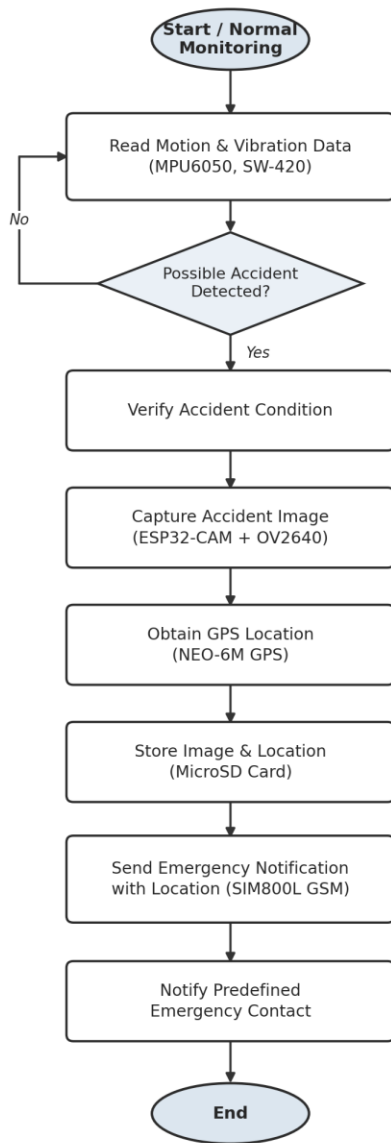


Figure 3: Flowchart of Accident Detection and Emergency Response

3.4 Driver/Rider Drowsiness Detection and Voice Alert

During normal riding, the ESP32-CAM periodically captures images of the rider's face to track eye state. Repeated detection of closed or half-closed eyes over a short time window is treated as an early sign of drowsiness, at which point the system immediately issues a voice warning through the small speaker and amplifier circuit. If the condition persists or worsens, the system repeats the alert at shorter intervals to keep the rider's attention. The voice output in this project is intentionally limited to safety-related alerts and system notifications — helmet, alcohol, speed, drowsiness, and theft warnings — rather than acting

as a general-purpose voice assistant, which keeps the feature realistic for a prototype-level microcontroller and amplifier circuit.

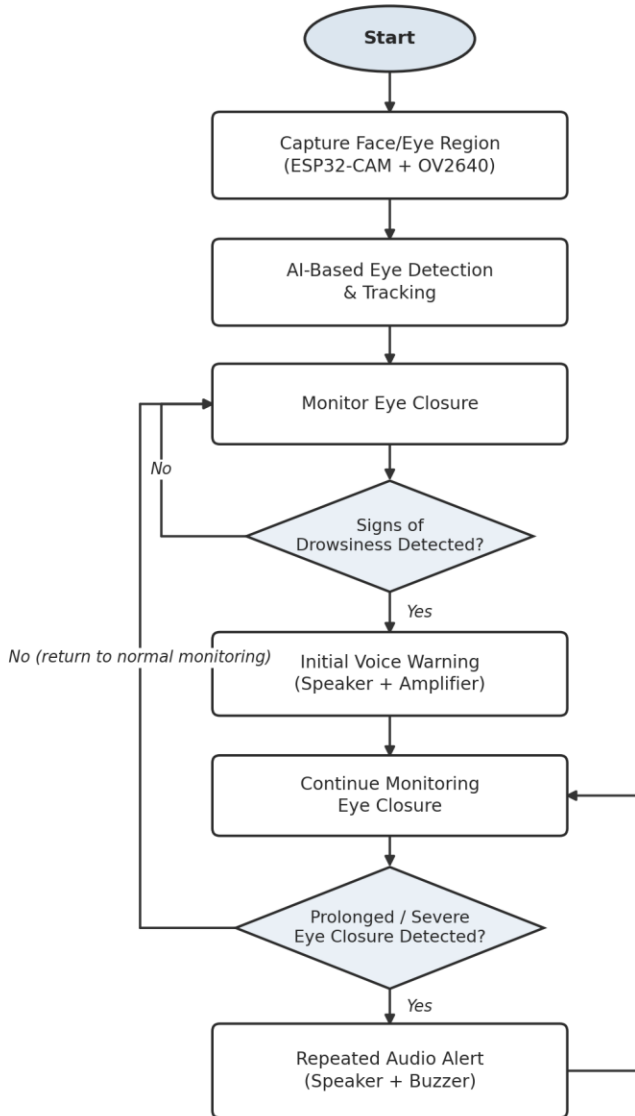


Figure 4: Flowchart of Driver Drowsiness Detection and Voice Alert

4. System Architecture and Working Principle

The ESP32 DevKit acts as the central processing and control unit of the system. It receives data from the camera, biometric, environmental, and motion sensors, applies the decision logic described in Section 3, and drives the output devices — the relay, buzzer, speaker, LEDs, and GSM module — accordingly. Figure 1 shows the overall block diagram of the system, grouped into an input/sensing layer, the central controller, an output/actuation layer, and a communication and logging layer.

The ESP32-CAM operates largely as a dedicated camera-processing node: it handles helmet detection and drowsiness monitoring and passes its results to the ESP32 DevKit, which coordinates the rest of the system. Separating the camera workload from the main controller keeps the central control loop responsive, since image capture and processing are comparatively slow operations.

4.1 Working Sequence

Although several checks in the system run continuously and in parallel once the motorcycle is moving, the logical sequence from power-on to emergency response follows the order shown in Figure 2. This

sequence reflects how the four feature groups from Section 3 connect to one another rather than a fixed, unchangeable execution order — in the actual firmware, speed, light, and drowsiness monitoring run as ongoing background checks rather than one-time steps.

In short: the system first confirms who is riding (fingerprint), then whether they are properly equipped and fit to ride (helmet and alcohol checks), before permitting ignition. Once moving, it watches speed, ambient light, and rider alertness, and separately watches for signs of a crash. If a crash is suspected, it gathers location data and notifies a predefined contact, while logging the event locally for later review.

5. Components Used in the Project

This section lists every hardware component selected for the prototype. For each component, the general-purpose function of the part is described first, followed by its specific role in this project and the key specifications relevant to that role. Prices and part-level electrical ratings are kept general where an exact figure was not confirmed, consistent with the reference material used for this section.

5.1 ESP32-CAM (with OV2640 Camera Module)



Figure 3: ESP32-CAM (with OV2640 Camera Module)

General Purpose: A compact Wi-Fi/Bluetooth-enabled microcontroller board with an integrated camera interface. The onboard OV2640 sensor captures still images or video frames that the board can process locally or forward to another controller.

Use in Our Project: Serves as the camera-based sensing node of the system. It captures the rider's face and posture for helmet detection, estimates whether more than one rider is present, monitors eye state for drowsiness detection, and can capture a still image at the moment an accident is suspected.

Key Specifications / Features:

- Microcontroller: ESP32 (dual-core, Wi-Fi + Bluetooth)

- Camera: OV2640, common resolutions up to 2MP
- Interfaces: UART for communication with the ESP32 DevKit
- Compact form factor suitable for helmet or handlebar mounting

5.2 ESP32 DevKit



Figure 4: ESP32 DevKit

General Purpose: A dual-core Espressif microcontroller board built around the ESP-WROOM-32 module, combining Wi-Fi and Bluetooth connectivity with a rich set of GPIO, ADC, and communication interfaces.

Use in Our Project: Functions as the central processing and control unit of the entire system. It reads data from all sensors (alcohol, speed, light, vibration, motion, GPS), runs the decision logic for each safety feature, and drives the relay, buzzer, speaker, and LED outputs.

Key Specifications / Features:

- Processor: Dual-core Tensilica Xtensa LX6, up to 240 MHz
- Memory: 520 KB SRAM, typically 4 MB external flash
- Wireless: Wi-Fi 802.11 b/g/n, Bluetooth 4.2 (BR/EDR & BLE)
- Interfaces: ADC, DAC, PWM, I2C, SPI, UART
- Power: 5V via USB or 4.8–12V via VIN, regulated to 3.3V onboard

5.3 MQ-3 Alcohol Sensor

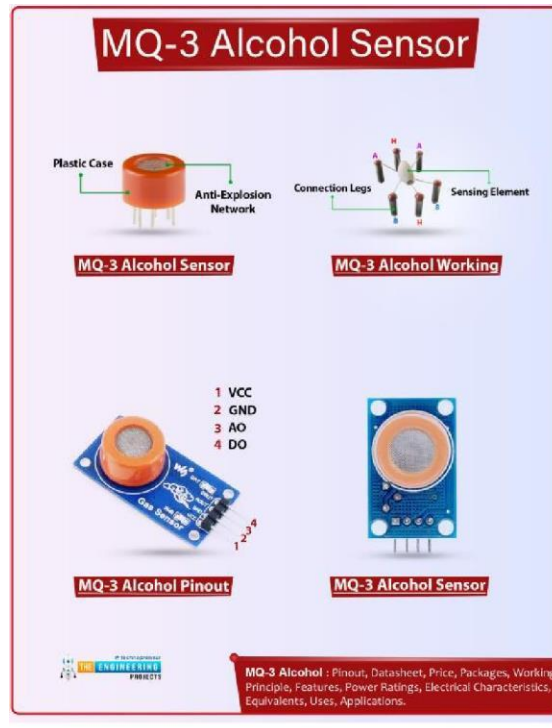


Figure 5: MQ-3 Alcohol Sensor

General Purpose: A semiconductor gas sensor whose electrical resistance changes in the presence of ethanol vapor, giving an analog voltage roughly proportional to alcohol concentration in the surrounding air.

Use in Our Project: Screens the rider for alcohol presence before the ignition circuit is allowed to activate. If the reading exceeds a configured threshold, the ESP32 keeps the relay open and triggers the buzzer warning.

Key Specifications / Features:

- Sensing element: Tin dioxide (SnO_2)
- Output: Analog voltage proportional to ethanol concentration
- Requires a brief warm-up period after power-up before readings stabilize
- Note: Not a certified/legally accurate breathalyzer — suitable only for threshold-based demonstration in a prototype

5.4 MPU6050 (6-Axis IMU)

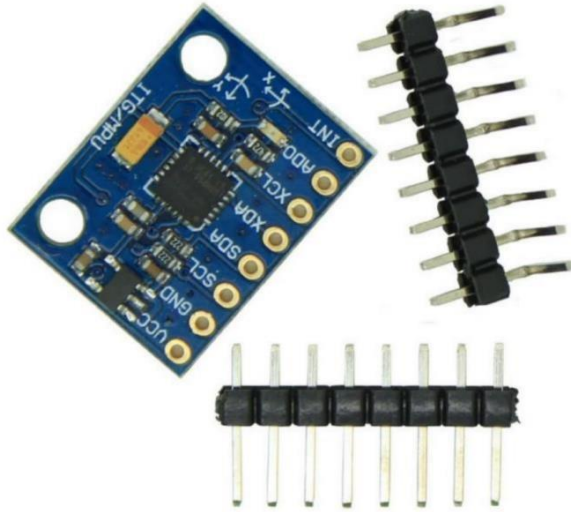


Figure 6: MPU6050 (6-Axis IMU)

General Purpose: A 6-axis inertial measurement unit combining a 3-axis accelerometer and 3-axis gyroscope on a single chip, communicating over I2C and offering onboard sensor fusion.

Use in Our Project: Continuously monitors the motorcycle's acceleration and orientation to help identify a sudden impact, abnormal tilt, or fall consistent with a possible accident, working together with the SW-420 vibration sensor.

Key Specifications / Features:

- Interface: I2C
- Accelerometer range: selectable $\pm 2g/\pm 4g/\pm 8g/\pm 16g$
- Gyroscope range: selectable ± 250 to $\pm 2000^\circ/s$
- Onboard Digital Motion Processor for sensor fusion

5.5 NEO-6M GPS Module

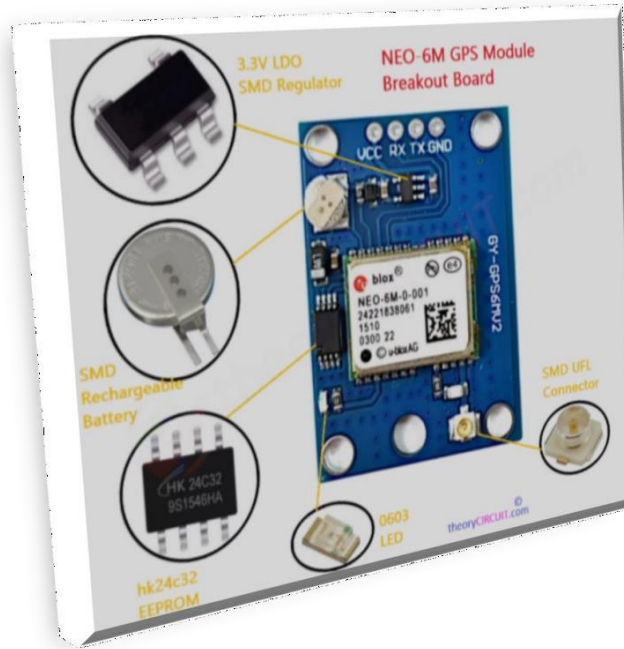


Figure 7: NEO-6M GPS Module

General Purpose: A u-blox-based satellite positioning module that communicates over UART and outputs standard NMEA sentences containing latitude, longitude, altitude, and time.

Use in Our Project: Provides the rider's current location once an accident is suspected. The coordinates it reports are combined with the SIM800L GSM module to compose an emergency SMS notification.

Key Specifications / Features:

- Interface: UART/TTL, default 9600 bps
- Output: Standard NMEA sentences (latitude, longitude, altitude, UTC time)
- Built-in ceramic patch antenna
- Compatible with 3.3V and 5V microcontrollers

5.6 Hall Effect Speed Sensor

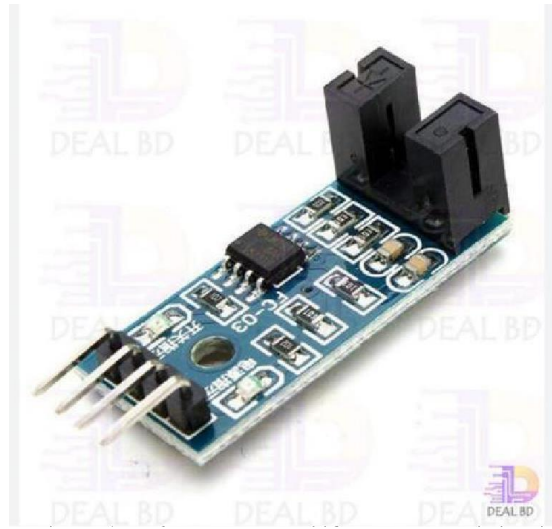


Figure 8: Hall Effect Speed Sensor

General Purpose: A magnetic sensor that converts wheel rotation into a digital pulse train; the pulse frequency corresponds directly to rotational (and therefore road) speed.

Use in Our Project: Mounted near the wheel, it lets the ESP32 calculate the motorcycle's approximate speed in real time and raise an overspeed warning when a rider-configured limit is exceeded.

Key Specifications / Features:

- Output: Digital pulse signal, one or more pulses per wheel rotation
- Speed derived from pulse frequency and wheel circumference
- Valued in automotive use for ruggedness compared to optical alternatives

5.7 SIM800L GSM Module

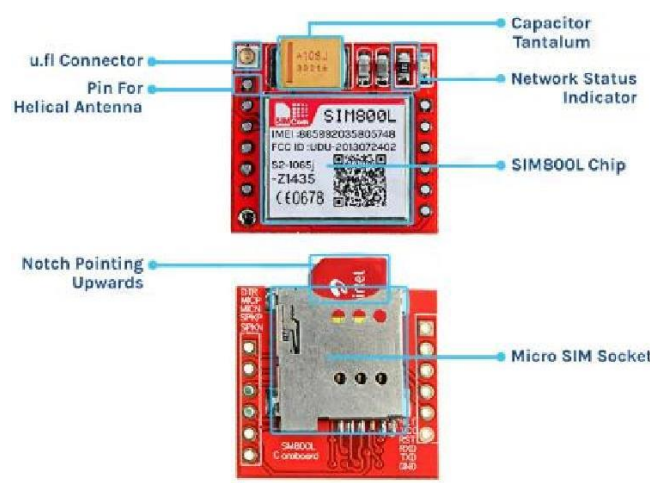


Figure 9: SIM800L GSM Module

General Purpose: A compact quad-band GSM/GPRS module controllable over UART using AT commands, capable of sending SMS, placing calls, and transferring data over GPRS.

Use in Our Project: Sends the emergency SMS notification (including the GPS coordinates) to a predefined contact when an accident is detected, and separately sends an anti-theft alert SMS when unauthorized access is attempted.

Key Specifications / Features:

- Interface: UART, controlled via AT commands
- Function used: SMS messaging to a stored emergency contact number
- Power: Requires a dedicated 3.7–4.2V supply capable of sustaining current spikes up to ~2A
- Note: Depends on 2G network availability from the local carrier

5.8 LDR (Light Dependent Resistor) Module

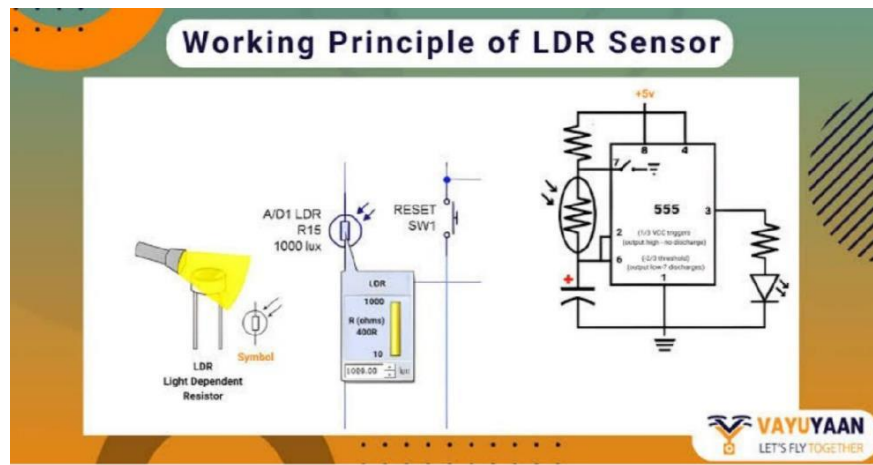


Figure 10: LDR (Light Dependent Resistor) Module

General Purpose: A passive light sensor whose resistance drops in bright light and rises in darkness, commonly wired into a voltage-divider circuit with a digital threshold output.

Use in Our Project: Monitors ambient light around the motorcycle. When the reading falls below a set threshold — for example, at dusk or when entering a tunnel — the ESP32 increases the brightness of the connected headlight/LED circuit.

Key Specifications / Features:

- Output: Digital HIGH/LOW via onboard comparator, with adjustable sensitivity
- Simple, low-power, low-cost light sensing
- Suitable for day/night or bright/dim threshold detection

5.9 Fingerprint Sensor



Figure 11: Fingerprint Sensor

General Purpose: A standalone biometric module (e.g., R307/R503/AS608 class) that captures, stores, and matches fingerprint templates onboard, without requiring the host controller to perform image processing.

Use in Our Project: Authenticates the rider before ignition is permitted. Only fingerprints previously enrolled in the module's onboard memory are accepted; repeated failed attempts are treated as a possible unauthorized-access event.

Key Specifications / Features:

- Interface: UART (TX/RX) to the ESP32
- Matching modes: 1:1 verification and 1:N identification
- Onboard template storage (capacity varies by model)
- Status feedback via LED indicator and/or serial response code

5.10 SW-420 Vibration Sensor

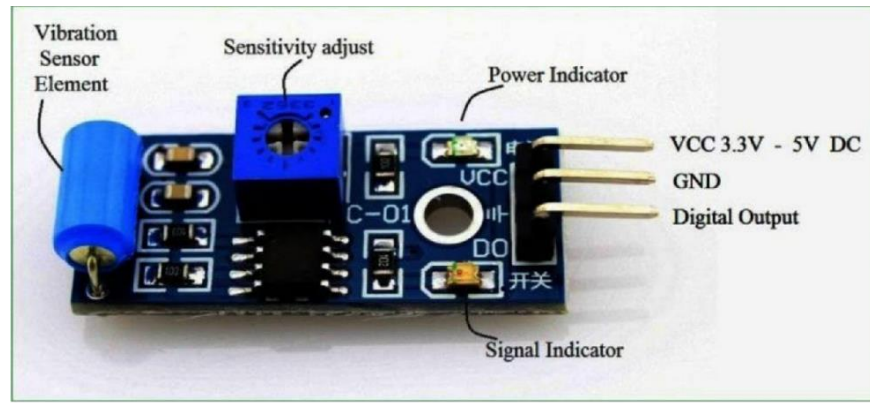


Figure 12: SW-420 Vibration Sensor

General Purpose: A mechanical vibration switch combined with an LM393 comparator and an adjustable sensitivity potentiometer, producing a digital HIGH signal when vibration exceeds a set threshold.

Use in Our Project: Provides a secondary, threshold-based indicator of a sudden impact or shock, used together with the MPU6050 to reduce false positives when deciding whether a probable accident has occurred.

Key Specifications / Features:

- Output: Digital HIGH/LOW via onboard comparator
- Adjustable sensitivity potentiometer
- Presence/absence detector — indicates that vibration occurred, not its magnitude or direction

5.11 Relay Module

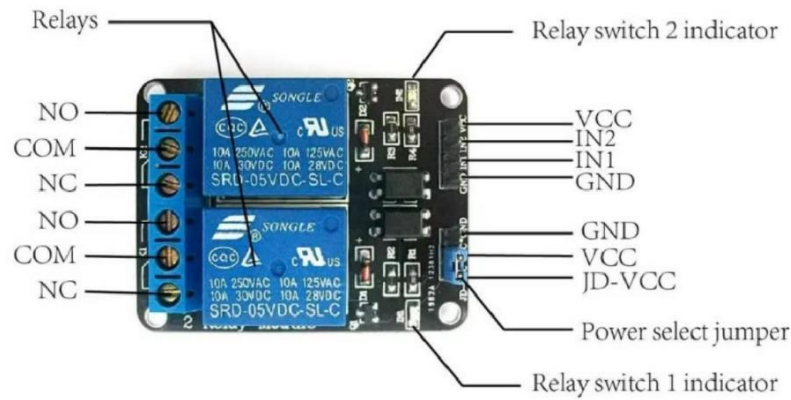


Figure 13: Relay Module

General Purpose: An electromagnetic switch that lets a low-voltage control signal from a microcontroller safely operate a higher-current circuit, with electrical isolation between the control and load sides.

Use in Our Project: Acts as the ignition-control element of the system. The ESP32 energizes the relay only after the helmet, alcohol, and authentication checks are satisfied, allowing the ignition circuit to close.

Key Specifications / Features:

- Control side: Digital signal from ESP32 GPIO (via driver circuit)
- Load side: Electrically isolated switching of the ignition circuit
- Includes onboard status LED and flyback protection in most modules

5.12 Buzzer



Figure 14: Buzzer

General Purpose: A small piezoelectric or electromagnetic component that converts an electrical signal into audible sound, commonly used for alerts and confirmation tones in embedded systems.

Use in Our Project: Provides immediate, local audible warnings for the alcohol interlock, overspeed condition, and unauthorized-access/anti-theft alerts, complementing the voice alerts issued through the speaker.

Key Specifications / Features:

- Type: Piezo (active or passive), 3-pin VCC/GND/Signal interface
- Active buzzers: single fixed tone, simple HIGH/LOW control
- Typical loudness: roughly 75–85 dB at close range for standard piezo modules

5.13 Small Speaker + Amplifier



Figure 15: Small Speaker + Amplifier

General Purpose: A small speaker paired with a low-power audio amplifier circuit, used to convert a weak electrical audio signal from a microcontroller or audio module into audible sound.

Use in Our Project: Delivers the system's voice alerts — drowsiness warnings, overspeed notices, alcohol-interlock warnings, and anti-theft alarm tones — as described in Section 3.4 and Section 3.2.

Key Specifications / Features:

- Function: Amplifies low-power audio signals for human-audible output
- Used for short, pre-defined safety voice prompts rather than open-ended speech
- Paired with the ESP32 or an audio playback module for stored voice clips

5.14 LEDs, Resistors, and Buttons



Figure 16: LEDs, Resistors, and Buttons

General Purpose: Basic discrete components: LEDs for visual indication, resistors for current-limiting and voltage-divider circuits, and push-buttons for manual input and configuration.

Use in Our Project: LEDs provide visual status indication (e.g., ignition ready, alert active) and supplement the automatic lighting/headlight-brightness feature; resistors support correct biasing of the LDR, buzzer, and LED circuits; buttons allow manual actions such as acknowledging an alert or entering a configuration mode during testing.

Key Specifications / Features:

- Standard through-hole LEDs and carbon-film resistors
- Momentary tactile push-buttons for manual input
- Used throughout the prototype circuit for indication and control

5.15 MicroSD Card Module

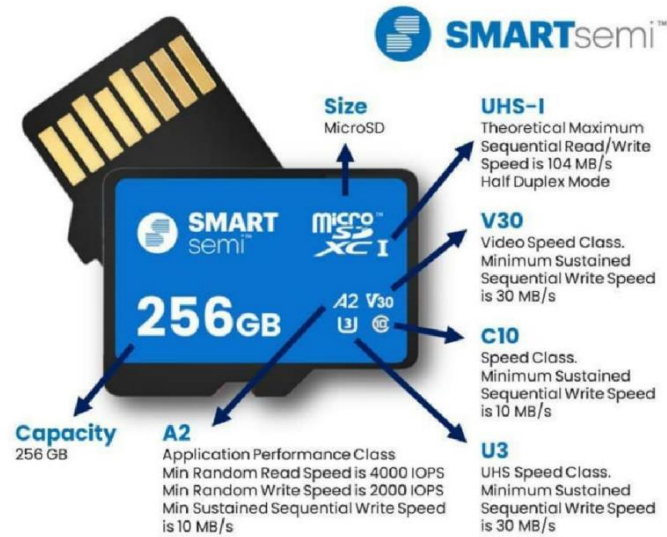


Figure 17: MicroSD Card Module

General Purpose: Allows a microcontroller to read from and write to a removable microSD card over SPI, extending the very limited onboard memory of the ESP32 for data storage.

Use in Our Project: Logs system events locally — for example, timestamps of ignition attempts, detected alerts, and accident events — providing a record that can be reviewed after the fact even without network connectivity.

Key Specifications / Features:

- Interface: SPI (MOSI, MISO, SCK, CS)
- Card format: FAT16/FAT32 for standard library compatibility
- Onboard 3.3V regulator/level shifting on most modules

5.16 5V/12V Converter / Regulator



Figure 18: 5V/12V Converter / Regulator

General Purpose: A voltage regulation module that steps down or steps up supply voltage to the levels required by different parts of a circuit.

Use in Our Project: Provides stable, appropriate supply voltages to the ESP32, sensors, and output modules from the motorcycle's battery or a bench supply during prototype development, protecting components from unsuitable voltage levels.

Key Specifications / Features:

- Provides regulated 5V and/or 12V rails as required by connected modules
- Supports the overall power budget of the sensor and actuator set
- Selected/sized according to the current draw of the GSM and camera modules, which are the largest consumers

5.17 Breadboard, Jumper Wires, and PCB/Connectors



Figure 19: Breadboard, Jumper Wires, and PCB/Connectors

General Purpose: Standard prototyping hardware: a solderless breadboard and jumper wires for quick circuit assembly and testing, and a PCB/perfboard with connectors for a more permanent build.

Use in Our Project: The breadboard and jumper wires are used during development to connect and test the ESP32, sensors, and modules without soldering. Once the design is verified, the PCB/perfboard and connectors are used to assemble a more stable final prototype circuit.

Key Specifications / Features:

- Breadboard: solderless prototyping board for initial testing
- Jumper wires: male-male/male-female for flexible connections
- PCB/perfboard and connectors: used for the final, more durable assembly

6. Feature-to-Component Integration Summary

Table 1 summarizes which hardware components are responsible for each system feature described in Section 3, providing a quick cross-reference between the functional description and the component details in Section 5.

System Feature	Components Involved
Helmet & Multi-Rider Detection	ESP32-CAM + OV2640, ESP32 DevKit, Relay Module
Alcohol Detection & Ignition Interlock	MQ-3 Sensor, ESP32 DevKit, Relay Module, Buzzer
Speed Monitoring & Overspeed Warning	Hall Effect Speed Sensor, ESP32 DevKit, Buzzer/Speaker
Automatic Low-Light Headlight Adjustment	LDR Module, ESP32 DevKit, LEDs
Rider Authentication	Fingerprint Sensor, ESP32 DevKit, Relay Module
Anti-Theft Alert	Fingerprint Sensor, ESP32 DevKit, Buzzer, SIM800L GSM
Accident Detection	MPU6050, SW-420, ESP32 DevKit, ESP32-CAM
GPS Location & Emergency Notification	NEO-6M GPS, SIM800L GSM, ESP32 DevKit
Drowsiness Detection & Voice Alert	ESP32-CAM, ESP32 DevKit, Speaker + Amplifier
Data Logging	MicroSD Card Module, ESP32 DevKit
Power Supply	5V/12V Converter/Regulator
Prototype Assembly	Breadboard, Jumper Wires, PCB/Connectors, LEDs/Resistors/Buttons

Table 1: Mapping between system features and hardware components.

7. Methodology

The project will be developed in the following stages:

- 1. Requirement Analysis:** Reviewing common causes of motorcycle accidents and theft to finalize the exact feature set and the safety thresholds (speed limit, alcohol threshold, drowsiness timing) appropriate for a prototype demonstration.
- 2. Hardware Selection and Procurement:** Sourcing the components listed in Section 5 and verifying their compatibility (voltage levels, communication interfaces) with the ESP32 DevKit and ESP32-CAM.

3. Circuit and System Design: Preparing the wiring/connection diagram for all sensors and actuators, allocating GPIO/ADC/UART/I2C/SPI pins on the ESP32 DevKit, and planning the power distribution through the 5V/12V regulator.

4. Individual Sensor Integration: Interfacing and testing each sensor (MQ-3, MPU6050, SW-420, Hall effect speed sensor, LDR, NEO-6M GPS) independently before combining them, to confirm each produces expected readings in isolation.

5. Camera and Image-Processing Implementation: Setting up the ESP32-CAM for helmet detection and drowsiness monitoring, and tuning the detection logic using sample images captured under different lighting conditions.

6. Authentication Module Implementation: Enrolling authorized fingerprints and implementing the verification logic that gates the ignition relay.

7. Communication Module Integration: Configuring the SIM800L GSM module for SMS alerts and linking it with the NEO-6M GPS output to compose location-based emergency messages.

8. Alert System Development: Implementing buzzer tones and short pre-recorded/synthesized voice prompts through the speaker and amplifier circuit for each alert condition.

9. Software and Hardware Integration: Combining all subsystems into the full firmware running on the ESP32 DevKit, implementing the decision logic and working sequence described in Section 4.

10. Testing and Calibration: Bench-testing each feature individually and then as a complete system, calibrating sensor thresholds (alcohol, vibration, speed, light) to reduce false positives and false negatives.

11. Final Prototype Assembly: Transferring the verified breadboard circuit to a PCB/perfboard, mounting components appropriately on a test motorcycle or mock-up frame, and performing an end-to-end demonstration.

12. Documentation and Presentation Preparation: Preparing the final report, circuit diagrams, and a live or recorded demonstration for evaluation.

8. Expected Outcomes

By the end of development, the prototype is expected to demonstrate the following capabilities under normal test conditions. As with any sensor-based prototype, none of these functions are expected to be perfectly or 100% accurate; the aim is to reliably demonstrate the intended behavior under reasonable test conditions rather than guarantee flawless detection in every scenario.

- Detect the presence or absence of a helmet on the rider using the onboard camera and prevent ignition when no helmet is detected.
- Detect an elevated alcohol reading from the MQ-3 sensor and withhold ignition accordingly.

- Authenticate a registered rider through the fingerprint sensor before permitting the motorcycle to start.
- Monitor and report approximate riding speed, issuing a warning when a configured limit is exceeded.
- Automatically adjust headlight/LED brightness in response to changing ambient light levels.
- Identify a probable accident event from combined motion and vibration data and respond by acquiring GPS coordinates and sending an SMS alert.
- Detect visible signs of drowsiness (prolonged eye closure) from the camera feed and issue a voice warning.
- Raise a local and remote (SMS) alert when an unauthorized fingerprint is used or repeated failed access attempts occur.
- Log key system events to the microSD card for later review.
- Demonstrate, as an integrated whole, a measurable improvement in rider safety awareness and a reduced expected response time to accidents compared to a motorcycle without such a system.

9. Limitations and Future Improvements

9.1 Limitations

- Camera-based helmet and drowsiness detection accuracy can degrade in poor lighting, heavy rain, or when the rider's face/head is partially obscured.
- The MQ-3 sensor is a low-cost semiconductor sensor and is not a certified breathalyzer; its readings are useful for threshold-based demonstration but are not legally or medically precise.
- Vibration- and motion-based accident detection can produce false positives (e.g., riding over a pothole) or false negatives (e.g., a low-speed tip-over), and thresholds will need careful calibration.
- GPS accuracy and acquisition time depend on satellite visibility and can be degraded in tunnels, dense urban areas, or under heavy cloud cover.
- GSM-based alerts depend on local network coverage and continued availability of 2G service, which is being phased out by some carriers.
- The prototype is built and tested at a small scale and may require additional ruggedization (weatherproof enclosures, vibration-resistant mounting) before real-world deployment on an actual motorcycle.

- Fingerprint sensor performance can be affected by wet, dirty, or gloved fingers, which is a realistic condition for motorcycle riders.

9.2 Future Improvements

- Replacing the MQ-3 with a fuel-cell or infrared-based alcohol sensor for more reliable readings.
- Adding a mobile application to let the owner view live status, past alerts, and location history rather than relying solely on SMS.
- Incorporating a rechargeable backup battery so the system continues to operate briefly if the main motorcycle battery is damaged in a crash.
- Extending the camera-based detection with a trained on-device model for more robust helmet and drowsiness classification across lighting conditions.
- Adding vehicle-to-cloud data logging (in addition to the local microSD log) once network conditions allow for more consistent connectivity.

10. Estimated

Table 2 gives an approximate cost estimate for the components required to build one prototype unit. These figures are estimated/approximate values in US dollars based on typical hobbyist-electronics pricing and are intended only to give a sense of overall prototype cost; actual prices will vary by supplier, region, and currency at the time of purchase.

Component	Qty
ESP32-CAM + OV2640	1
ESP32 DevKit	1
MQ-3 Alcohol Sensor	1
MPU6050	1
NEO-6M GPS Module	1
Hall Effect Speed Sensor	1
SIM800L GSM Module	1
LDR Module	1
Fingerprint Sensor	1
SW-420 Vibration Sensor	1

Component	Qty
Relay Module	1
Buzzer	2
Small Speaker + Amplifier	1
LEDs + Resistors + Buttons (assorted)	Set
MicroSD Card Module + Card	1
5V/12V Converter/Regulator	1
Breadboard + Jumper Wires + PCB/Connectors	Set
Miscellaneous (wiring, enclosure, mounting hardware)	—
Estimated Total	

Table 2: Approximate/estimated component cost for one prototype unit.

12. References

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